ADITHYA B K

Software Engineer | Full-Stack Development
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Mandya, India

Computer Science undergraduate with hands-on experience building full-stack, production-style web platforms and applied ML/NLP integrations. Skilled in Python, React.js, Next.js, TypeScript, and FastAPI, with a strong foundation in data structures, algorithms, and system design. Comfortable working across the stack — from REST API design, authentication and real-time features to responsive UI and database architecture.

EDUCATION

PES College of Engineering, Mandya Oct 2023 – Oct 2027
B.E. in Computer Science and Engineering — Artificial Intelligence & Machine Learning, currently pursuing 4th year, with 7.73 CGPA

EXPERIENCE

Machine Learning Trainee — Internship, Employability.life Aug 2024 – Oct 2024
- Built a supervised-learning classification pipeline on a synthetic dataset, covering preprocessing, model training, and evaluation end to end.
- Applied correlation analysis to reduce feature dimensionality (32 to 18) and validated model performance using 5-fold cross-validation. Technologies: Python, Scikit-learn, Pandas.

PROJECTS

Kisan Sethu — Agentic AgriTech Marketplace — Full-Stack Development
- Architected a peer-to-peer marketplace for farmers, buyers, field agents and admins with 33 REST endpoints, JWT authentication and role-based access control across four user types.
- Built an autonomous reverse-auction bidding loop as a FastAPI background task, Razorpay-sandbox payments with delivery-OTP verification, shipment tracking and role-specific dashboards.
- Integrated Gemini Vision for AI crop-quality grading, plus demand forecasting charts and multilingual voice onboarding.

Tech Stack: Next.js, React, TypeScript, FastAPI, SQLAlchemy, SQLite, Tailwind CSS, Gemini API

RecoverAI — AI Revenue Recovery Agent (Razorpay Buildathon, Track 03) — Product & Frontend Engineering
- Designed a 6-stage decision pipeline (risk detection, diagnosis, recommendation, 9-rule policy and safety gate, simulation, audit trail) over a deterministic 1,000-record synthetic payment dataset.
- Built the merchant dashboard with a transaction investigation drawer and one-click demo scenarios; every decision is logged with an audit ID persisted in localStorage.

Tech Stack: React 19, TypeScript, Vite, Tailwind CSS 4

SmartPave — AI-Based Pavement Maintenance Optimization — Full-Stack & ML
- Built a FastAPI + scikit-learn Random Forest service predicting pavement condition (PCI) for 12 road sections, with life-cycle-cost analysis and a React dashboard for decision support.

Tech Stack: Python, FastAPI, scikit-learn, React, TypeScript, Recharts

MediRead — AI-Based Medical Report Understanding System — Backend & AI Integration
- Built a pipeline to extract lab values from PDF medical reports using text parsing and regex, with rule-based abnormal-value detection.
- Integrated Llama3 (via Ollama) for multilingual explanations with voice output; deployed a web version on Vercel.

Tech Stack: Python, NLP, Llama3/Ollama, React, Vercel

Startup Investment Trends in India — Data Analytics & BI — Backend Data Pipeline
- Built a Python and MySQL pipeline to clean and query startup funding data, with interactive Power BI dashboards. Tech Stack: Python, MySQL, Power BI.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL, Swift
Frameworks & Libraries: React.js, Next.js, FastAPI, Tailwind CSS, LangChain, scikit-learn
Databases: PostgreSQL, MySQL, SQLite, Firebase
Core CS: Data Structures & Algorithms, REST API Design, System Design, Authentication & Authorization
Tools & Platforms: Git, GitHub, VS Code, Linux, Vercel, Judge0 API

CERTIFICATIONS

- Introduction to DevOps — IBM
- Software Engineering: Software Design and Project Management — HKUST
- Prompt Engineering for ChatGPT — Vanderbilt
- Deloitte Australia Technology Job Simulation — Forage

ADDITIONAL INFORMATION

- Qualified for Smart India Hackathon (SIH) 2025 at the national level.
- Runner-Up, Minute to Win (ISTE) — placed 2nd of 50+ participants.
- Industry 4.0 Foundation Certification — Microsoft & SAP CSR Initiative (TechSaksham / Edunet Foundation / AICTE).